

HALL A MOLLER DAQ FIRMWARE for FADC-250's Xilinx FX20 FPGA

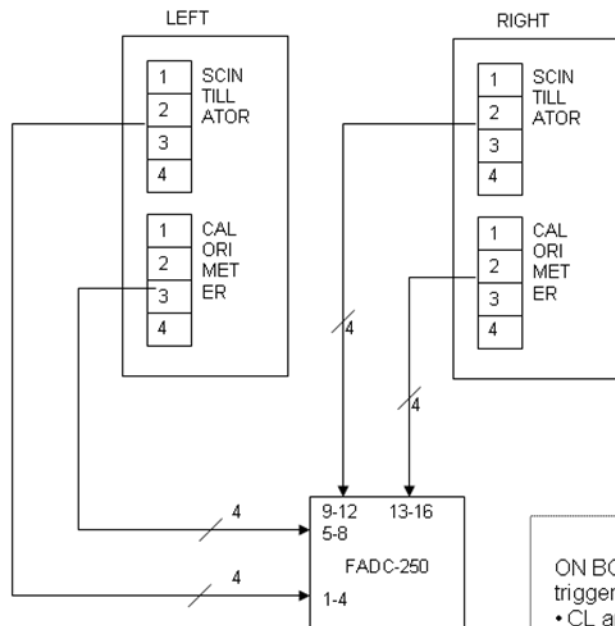
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MOLLER A Functional Description:

HALL A Moller Polarimeter Requirement



When TRIGGER condition is met, send data that cause TRIGGER.

TRIGGER condition (or):

CL AND CR prescaled from 1 to at least 1000

CL prescaled from 1 to at least 1000

CR prescaled from 1 to at least 1000

$$CR = \sum_{i=1,4} \sum_{j=1,2} P_i^j > \text{threshold}$$

$$CL = \sum_{i=1,4} \sum_{j=1,2} P_i^j > \text{threshold}$$

$$SL = (\sum_{j=1,2} S1^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S2^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S3^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S4^j > \text{threshold})$$

$$SR = (\sum_{j=1,2} S5^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S6^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S7^j > \text{threshold}) \text{ or } (\sum_{j=1,2} S8^j > \text{threshold})$$

ON BOARD SCALER (COUNTER) to be read out by a separate trigger (helicity and gate bits) at the helicity cycle of 30 to 2kHz.

- CL and CR
- CL and SL
- CR and SR
- CL and CR and SL and SR
- CL and CR and (SL and SR delayed > 100 ns)

Overview:

MOLLER DAQ FIRMWARE ARCHITECTURE Xilinx FX20

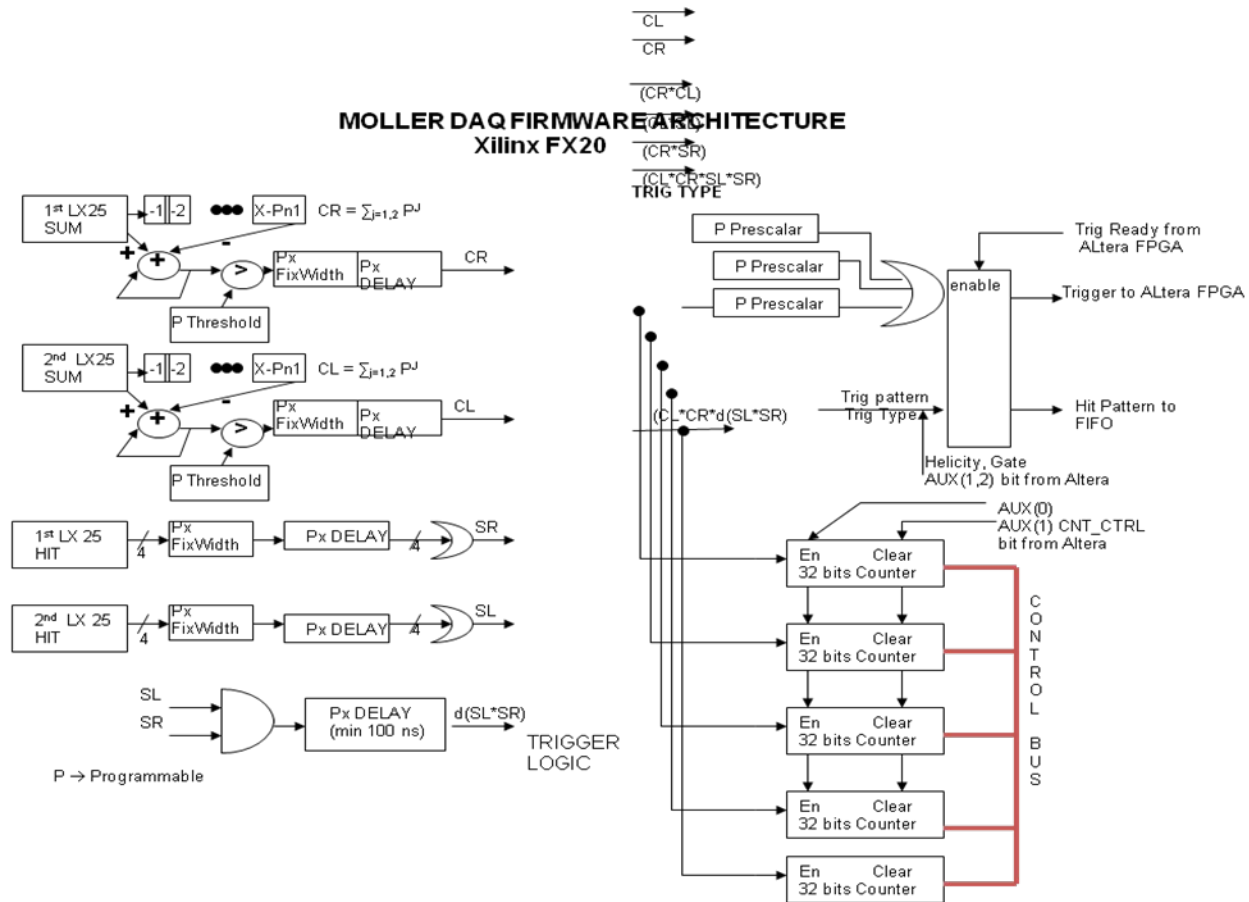


Figure 12: Control Bus Address Map
Do not use burst mode.

Name	Width (bits)	Access	Primary Address (hex)	Default Value (hex)	Function
STATUS0	16	R	400		x"08" & Version #
STATUS1_ADR	16	R	401		x"0000"
CLandCR_LO_COUNT	16	R	402		Low word of CL and CR Scaler
CLandCR_HI_COUNT	16	R	403		Hi word of CL and CR Scaler
CLandSL_LO_COUNT	16	R	404		Low word of CL and SL Scaler
CLandSL_HI_COUNT	16	R	405		Hi word of CL and SL Scaler
CRandSR_LO_COUNT	16	R	406		Low word of CR and SR Scaler
CRandSR_HI_COUNT	16	R	407		Hi word of CR and SR Scaler
CLaCRaSLaSR_LO_COUNT	16	R	408		Low word of CL and CR and SL and SR Scaler
CLaCRaSLaSR_HI_COUNT	16	R	409		Hi word of CL and CR and SL and SR Scaler
CLaCRaDelaySLaSR_HI_COUNT	16	R	40A		Low word of CL and CR and delay(SL and SR) Scaler
CLaCRaDelaySLaSR_LO_COUNT	16	R	40B		Hi word of CL and CR and delay(SL and SR) Scaler
CL_ADD_LENGTH	6	R/W	40C	3	Number of CL sample to add, Min is 3.
CL_SUM_LO_THREDSHOLD	16	R/W	40D	FF	Bit 15-0 of CL SUM threshold
CL_SUM_HI_THREDSHOLD	4	R/W	40E	0	Bit 20-16 of CL SUM threshold
CR_ADD_LENGTH	6	R/W	40F	3	Number of CR sample to add, Min is 3
CR_SUM_LO_THREDSHOLD	16	R/W	410	FF	Bit 15-0 of CR SUM threshold
CR_SUM_HI_THREDSHOLD	4	R/W	411	0	Bit 20-16 of CR SUM threshold
CL_PULSE_LENGTH	16	R/W	412	0	Set the pulse length of CL after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period.
CR_PULSE_LENGTH	16	R/W	413	0	Set the pulse length of CL after crossing threshold.

					Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period.
SL1_PULSE_LENGTH	16	R/W	414	0	Set the pulse length of SL1 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
SL2_PULSE_LENGTH	16	R/W	415	0	Set the pulse length of SL2 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting +1) Clock period
SL3_PULSE_LENGTH	16	R/W	416	0	Set the pulse length of SL3 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
SL4_PULSE_LENGTH	16	R/W	417	0	Set the pulse length of SL4 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
SR1_PULSE_LENGTH	16	R/W	418	0	Set the pulse length of SR1 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
SR2_PULSE_LENGTH	16	R/W	419	0	Set the pulse length of SR2 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
SR3_PULSE_LENGTH	16	R/W	41A	0	Set the pulse length of SR3 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock

					period
SR4_PULSE_LENGTH	16	R/W	41B	0	Set the pulse length of SR4 after crossing threshold. Setting of 0 give pulse width of 1 Clock period. Any other settings give pulse width of (setting + 1) Clock period
CL_DELAY	8	R/W	41C	0	Set the delay of CL pulse. Delay is (setting + 1) clock
CR_DELAY	8	R/W	41D	0	Set the delay of CR pulse. Delay is (setting + 1) clock
SL1_DELAY	8	R/W	41E	0	Set the delay of SL1 pulse. Delay is (setting + 1) clock
SL2_DELAY	8	R/W	41F	0	Set the delay of SL2 pulse. Delay is (setting + 1) clock
SL3_DELAY	8	R/W	420	0	Set the delay of SL3 pulse. Delay is (setting + 1) clock
SL4_DELAY	8	R/W	421	0	Set the delay of SL4 pulse. Delay is (setting + 1) clock
SR1_DELAY	8	R/W	422	0	Set the delay of SR1 pulse. Delay is (setting + 1) clock
SR2_DELAY	8	R/W	423	0	Set the delay of SR2 pulse. Delay is (setting + 1) clock
SR3_DELAY	8	R/W	424	0	Set the delay of SR3 pulse. Delay is (setting + 1) clock
SR4_DELAY	8	R/W	425	0	Set the delay of SR4 pulse. Delay is (setting + 1) clock
SLandSR_DELAY	8	R/W	426	0	Set the delay of SLandSR pulse. Delay is (setting + 1) clock
CR SCALAR	11	R/W	427	1	Set CR SCALAR. Zero is invalid
CL SCALAR	11	R/W	428	1	Set CL SCALAR Zero is invalid
CRandCL SCALAR	11	R/W	429	1	Set CRandCL SCALAR. Zero is invalid

Hit FIFO	16	R	42A		Read Hit Pattern
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Signal Mapping

SUM 1 => **CR** => **U93** ➔ **ADC(J) 13- 16**

HITSBITS1 (0-3) => **SR (0-3)** => **U93** ➔ **ADC(J) 9 - 12**

SUM 2 => **CL** => **U100** ➔ **ADC(J) 5 - 8**

HITSBITS2 (0-3) => **SL (0-3)** => **U100** ➔ **ADC(J) 1 - 4**

Delay (# of Clock):

HITSBITS to Px FixWidth IN => 4

Px FixWidth IN to Px FixWidth OUT => 1

NOTE #1: For Trigger Mode (Configuration bit 1,0 = 10), delay time from end of window to beginning of Live Trigger high.

